LED Driver PCB — Component Analysis

Reverse-engineering analysis based on the photographed PCB and the structure of the supplied Reverse Engineering Report.

1. Component Analysis

Ref.	Component	Identification / Marking	Electrical Role	Functional Role	Connected To
J1 / Input	2-pin screw terminal	Green terminal block	AC input connection	Receives electrical power from the mains	Input → Rectifier section
J2 / Output	2-pin screw terminal	Green terminal block	DC output connection	Supplies regulated DC power to LED strip	Output filter → LED load
T1	High-frequency transformer	Yellow ferrite transformer	Magnetic energy transfer and isolation	Transfers energy from primary to secondary while providing electrical isolation	Switching transistor ↔ Output rectifier
D4–D7	Diodes	Four individual diodes	AC-to-DC rectification	Convert incoming AC into pulsating DC	Input → DC bus
C1	Electrolytic capacitor	Large electrolytic capacitor	DC filtering/storage	Smooths rectified DC and stores energy	Rectifier → Switching stage
C5	Electrolytic capacitor	Large electrolytic capacitor	Output filtering	Reduces output voltage ripple and provides smoother DC	Secondary rectifier → Output
C3/C4	Capacitors	Green/blue capacitors	Filtering / suppression	Reduce electrical noise and switching spikes	Switching/primary circuit
R1–R6	Resistors	Colour-banded resistors	Current limiting, biasing and sensing	Set operating conditions and limit/control current	Switching and control network
Q1	High-voltage transistor	Marking appears as 13003D	High-frequency switching	Rapidly switches current through transformer primary	DC bus → Transformer primary
Q2	Transistor	TO-type package	Switching/control	Assists switching/control operation	Control network ↔ Switching stage
D1–D3	Diodes	Individual axial diodes	Rectification/clamping	Provide directional current paths and suppress unwanted voltage spikes	Primary/control section
Electrolytic capacitors	Polarized capacitors	Black/red cylindrical components	Energy storage and filtering	Smooth DC and stabilize supply/output	Rectifier and output stages
Film/ceramic capacitors	Non-polarized capacitors	Green/blue components	High-frequency filtering	Suppress switching noise and transients	Switching section
PCB	Printed Circuit Board	Yellow PCB	Electrical interconnection	Provides tracks and mechanical mounting for components	Connects complete circuit
Enclosure	Plastic housing	EMAFORO LED DRIVER	Protection/interface	Protects circuit and provides accessible terminals	Contains PCB

2. Detailed Component Explanations

2.1 Input Terminal

The 2-pin green screw terminal is the electrical input interface. It provides the connection between the external power source and the PCB. Function: it brings the input AC supply into the driver circuit.

2.2 Bridge Rectifier — D4 to D7

The four diodes at one side of the PCB form a bridge rectifier. Function: they convert the incoming AC voltage into pulsating DC voltage. Flow: AC Input → D4–D7 Bridge Rectifier → C1 → Switching Stage.

2.3 C1 — Input Electrolytic Capacitor

C1 stores electrical energy and filters the rectified supply. After rectification, the voltage is pulsating DC; C1 smooths this voltage and provides a relatively stable DC supply to the switching circuit.

2.4 Q1 — 13003D Switching Transistor

The component marked 13003D is a high-voltage switching transistor. Electrical role: high-frequency electronic switching. It rapidly switches current through the primary winding of transformer T1, converting DC energy into high-frequency pulses.

2.5 T1 — High-Frequency Transformer

T1 is the yellow transformer. Electrical role: energy transfer and electrical isolation. It transfers energy from primary to secondary, provides isolation between input and output, and allows the output voltage to be suitable for the LED load. It operates at high frequency, allowing a smaller transformer than a conventional 50-Hz transformer.

2.6 D1–D3 — Diodes

The smaller axial diodes are used in the primary/control network. Possible roles include rectification, current-path control, voltage clamping and protection against switching transients. The exact individual function of D1, D2 and D3 cannot be confirmed from the photographs alone; PCB track tracing or a schematic would be required.

2.7 C3/C4 — Small Capacitors

The green/blue capacitors are non-polarized capacitors used for filtering and noise suppression. They help suppress high-frequency noise and switching transients produced by the SMPS.

2.8 R1–R6 — Resistors

The colour-banded components are resistors. Their exact values can be determined using colour codes or multimeter measurements. Possible roles include current limiting, transistor biasing, voltage sensing, startup/control current, capacitor discharge and switching-current limiting. Exact individual functions should be confirmed by tracing PCB tracks.

2.9 C5 — Output Electrolytic Capacitor

C5 performs output filtering. It reduces ripple in the rectified secondary voltage and produces smoother DC output for the LED load. Flow: T1 secondary → Output rectification → C5 → 12 V DC output.

2.10 Q2 — Transistor

Q2 is part of the switching/control network. Electrical role: switching/control. It helps control the operation of the switching stage and establishes required transistor bias/control conditions. Its exact type and precise function cannot be reliably identified from the photographs alone.

2.11 Output Terminal

The second 2-pin green screw terminal is the output interface. The enclosure specifies OUTPUT: DC/12V, so this terminal provides DC supply to the LED strip/load.

3. Overall Functional Architecture

AC Input
↓
Bridge Rectifier (D4–D7)
↓
Input Filter Capacitor (C1)
↓
High-Frequency Switching Transistor (Q1)
↓
High-Frequency Transformer (T1)
↓
Secondary Rectification
↓
Output Filter Capacitor (C5)

↓
12 V DC Output
↓
LED Strip

4. Reverse Engineering Conclusion

The PCB should not be viewed merely as a collection of resistors, capacitors and transistors. Its components form a power-conversion system: the input AC is rectified, filtered, switched at high frequency, transferred through an isolated transformer, rectified and filtered again to obtain the required DC output. This follows the reverse-engineering progression of component identification → specification → functional understanding → engineering reasoning.

5. Verification Note

Exact values for R1–R6, C3/C4, D1–D3 and Q2 should not be entered unless verified from component markings, datasheets, PCB track tracing or measurement. The photographs alone do not provide enough evidence to confirm every individual value or function.